Dietary Habits & Obesity Analysis

Latin American Population

Descriptive & Diagnostic Analysis

2,111	17	3
Participants	Variables	Countries

Dataset	UCI Machine Learning Repository
Countries	Mexico, Peru & Colombia
License	CC BY 4.0
Tools	Microsoft Excel · Power BI Desktop
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Date	May 2025

1. Introduction

This report presents a descriptive and diagnostic analysis of dietary habits and obesity levels in a Latin American population. The dataset was sourced from the UCI Machine Learning Repository and contains survey data from participants in Mexico, Peru, and Colombia, supplemented with synthetically generated data to balance class distribution.

This project was developed as part of a data analytics portfolio. The analysis applies data cleaning, exploratory data analysis, and dashboard visualization skills using Microsoft Excel and Power BI.

2. Research Questions

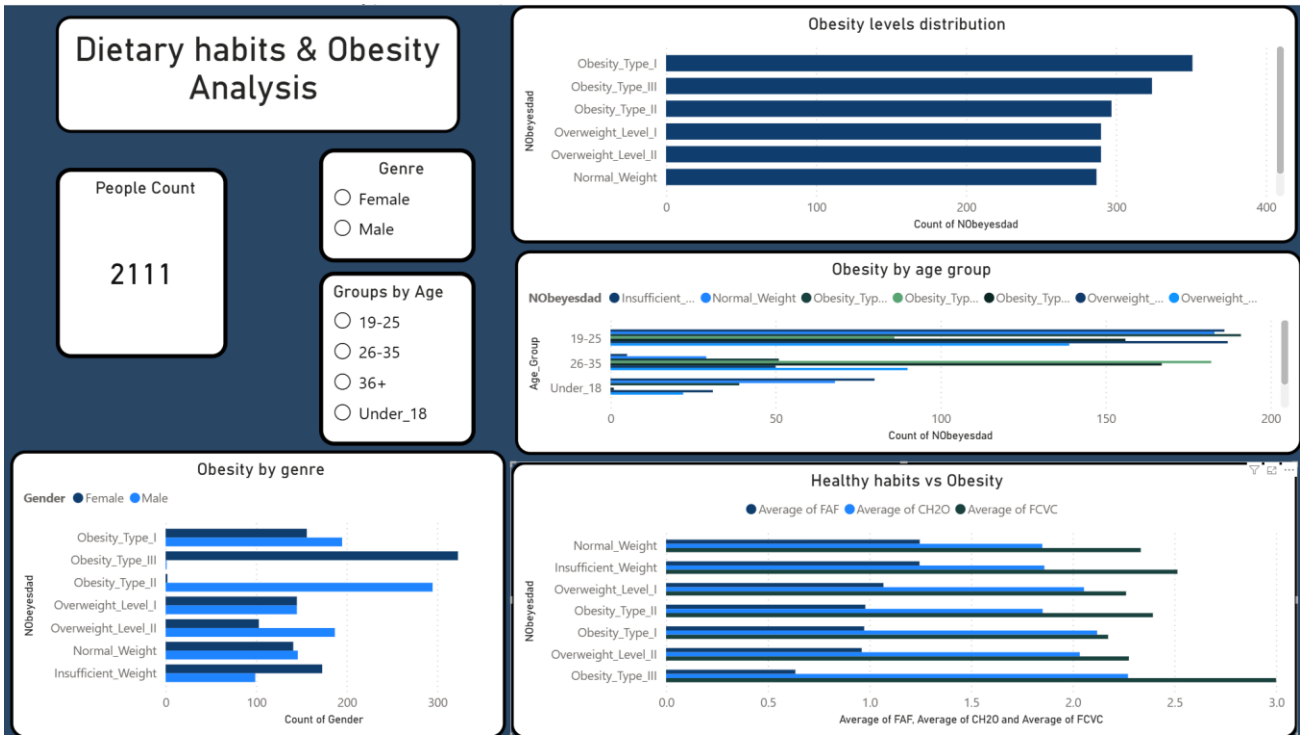
- Which dietary and lifestyle habits are most associated with higher obesity levels?
- Are there significant differences in obesity distribution between genders?
- How does obesity prevalence vary across age groups?
- What percentage of the population meets basic healthy habit thresholds for water intake, vegetable consumption, and physical activity?

3. Dataset Overview

The dataset contains 2,111 records and 17 original variables. Two additional columns were engineered during the data cleaning process: BMI and Age_Group.

Variable	Description	Type
Gender	Biological sex of participant	Categorical
Age	Age in years	Numeric
Height / Weight	Meters and kilograms	Numeric
FAVC	Frequent consumption of high-caloric food (Yes/No)	Categorical
FCVC	Frequency of vegetable consumption (scale 1-3)	Ordinal
NCP	Number of main meals per day (1-4)	Ordinal
CH2O	Daily water intake (1=<1L, 2=1-2L, 3=>2L)	Ordinal
FAF	Physical activity frequency (scale 0-3)	Ordinal
TUE	Daily screen time in hours (scale 0-2)	Ordinal
CALC	Alcohol consumption frequency	Categorical
MTRANS	Primary transportation method	Categorical
NObeyesdad	Obesity level — 7 categories	Target
BMI *	Engineered: Weight / Height squared	Derived
Age_Group *	Engineered: Under_18, 19-25, 26-35, 36+	Derived

** Columns created during the data cleaning process.*



4. Data Cleaning Process

The raw CSV file was loaded into Microsoft Excel. The original file was preserved in a /raw folder without modifications. All cleaning steps were applied to a working copy saved as obesity_clean.xlsx.

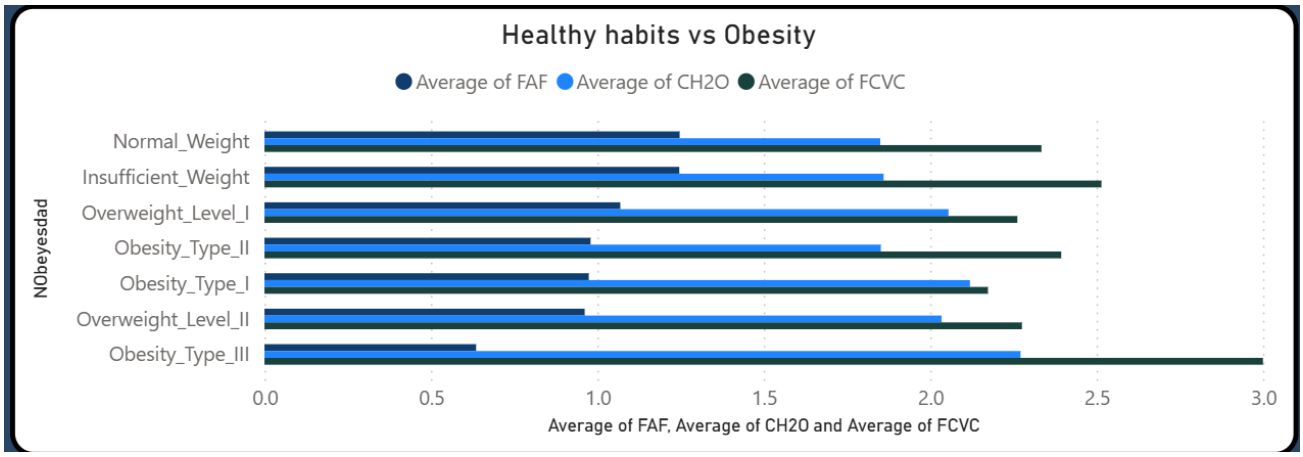
Synthetic Data Identification	77% of records were synthetically generated using the SMOTE algorithm. Evidence: numeric columns such as Age, Height, and FCVC contained decimal values impossible in a real survey (e.g., FCVC = 2.341). This was documented as a dataset characteristic, not treated as an error.
Ordinal Scale Rounding	FCVC, NCP, CH2O, FAF, and TUE were rounded to integer using =ROUND(cell, 0) to restore their original ordinal scale. Auxiliary columns were used to verify results before overwriting original values.
BMI Calculation	A new column BMI was added using =Weight/(Height*Height). This standard clinical metric enables comparison across participants independent of height.
Age Group Segmentation	A new column Age_Group was created using nested IF logic, segmenting participants into: Under_18, 19-25, 26-35, and 36+.
Null Value Verification	No null or empty values were found across all 17 columns. This was confirmed before and after all cleaning steps.

5. Key Findings

The following findings are based on direct observation of the Power BI dashboard visualizations and interpreted using nutritional domain knowledge.

Finding 1 — Physical Activity & Obesity Level

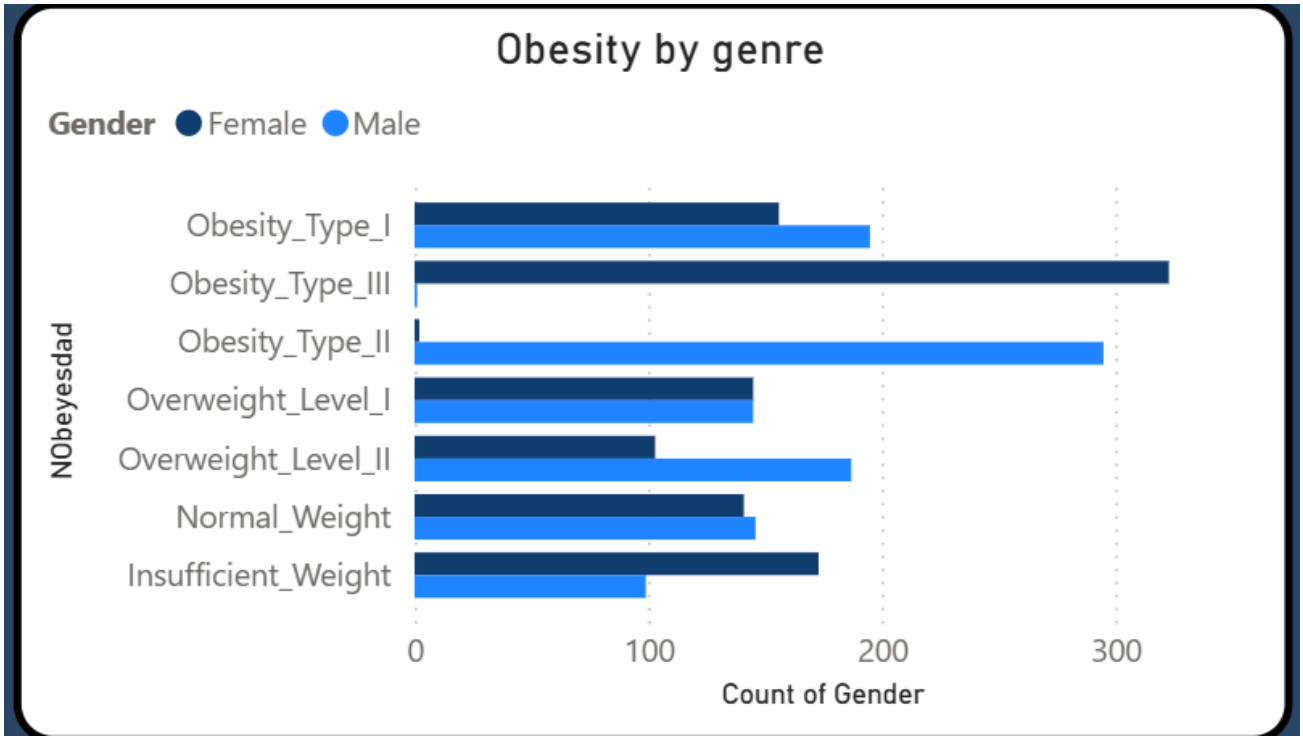
The Obesity_Type_III group shows the lowest physical activity average (FAF) across the entire sample. This is directly related to their current health status: the higher the obesity level, the lower the recorded physical activity. Conversely, participants closer to Normal_Weight demonstrate higher FAF averages. This suggests a clear inverse relationship between physical activity frequency and obesity level. The more active a person is, the closer they tend to be to a healthy weight.



Recommendation: Public health interventions targeting sedentary populations should prioritize increasing physical activity as a primary strategy, particularly for groups classified as Obesity_Type_I and above.

Finding 2 — Gender Distribution Across Obesity Types

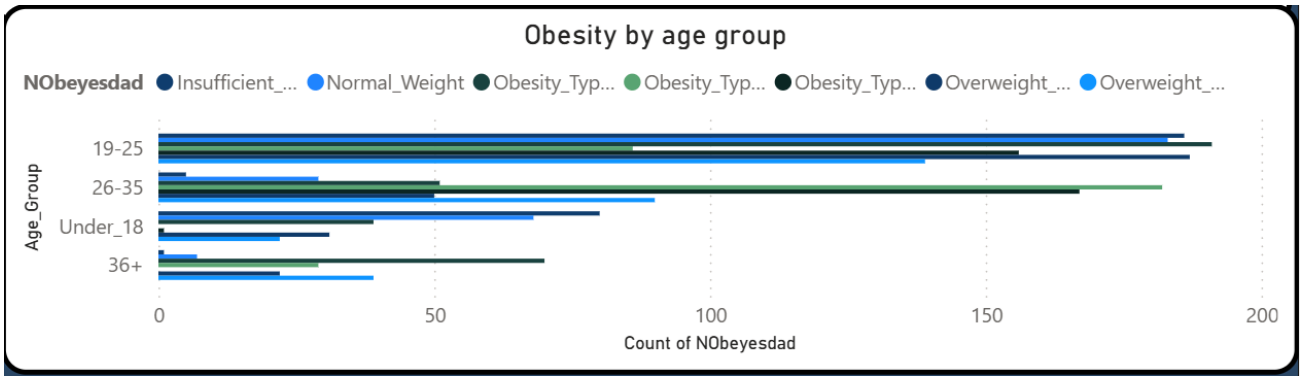
The most notable gender difference appears in Obesity_Type_III and Obesity_Type_II. The Obesity_Type_III group is represented almost entirely by women, while Obesity_Type_II shows the opposite, nearly 100% male, with only 2 female cases recorded. Beyond reflecting a possible biological pattern in fat distribution between genders, these differences are too large to be explained by biology alone. This likely indicates an error or bias in the data collection process, and the findings should be interpreted with caution.



Note: Real-world clinical data would be required to validate whether this gender pattern reflects an actual population trend or is an artifact of the synthetic data generation process.

Finding 3 — Obesity by Age Group

The Under_18 group presents a total of 40 people in an obese state, while the 36+ group records 99 people with the same condition. This suggests that at younger ages, higher physical activity levels contribute to better weight control. However, in the 36+ population, several additional factors may affect the results: reduced interest in maintaining a healthy weight, less balanced dietary patterns, and — importantly from a clinical standpoint — the decrease in basal metabolic rate that naturally occurs with age. This physiological change means the body burns fewer calories at rest, which can contribute to weight gain even without changes in diet or activity level.



6. Study Limitations

- 77% of records were synthetically generated (SMOTE), which may produce unrealistic correlations.
- Ordinal scales (1-3) limit measurement precision compared to continuous clinical measurements.
- The sample is biased toward young university students (53% aged 19-25).
- NObeyesdad classification does not directly correspond to standard WHO BMI ranges.
- Self-reported dietary data tends to overestimate healthy behaviors.
- The 36+ age group contains only 168 participants, limiting statistical power for that segment.

7. Conclusions

Physical activity frequency is the variable most consistently associated with healthy weight maintenance in this Latin American sample. Participants classified as Normal_Weight show higher averages in physical activity, water intake, and vegetables consumption compared to obese groups, suggesting that a combination of exercise, hydration, and vegetable consumption acts as a protective cluster against obesity.

The gender-based disparity in extreme obesity types warrants further investigation with real-world clinical data. The current findings are likely influenced by the synthetic data generation process.

Future analyses using real clinical datasets — including measured caloric intake, hormonal factors, and socioeconomic variables — would significantly strengthen the validity of these findings.

8. Technical Details

Data Source	UCI ML Repository — Estimation of Obesity Levels Based on Eating Habits
Dataset URL	archive.ics.uci.edu/dataset/544
License	CC BY 4.0
Raw Records	2,111 rows x 17 columns
Clean Records	2,111 rows x 19 columns (+ BMI, Age_Group)
Cleaning Tool	Microsoft Excel
Visualization Tool	Power BI Desktop
Project Files	obesity_clean.xlsx | obesity_study.pbix